

A Report
On
AI-ASSISTED MARINE RISK INTELLIGENCE &
COMMUNITY
REPORTING PLATFORM

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement for the award of degree of
Bachelor of Technology in CSE



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CANDIDATE'S DECLARATION

I, Ayush Kumar, do hereby declare that the internship report titled “AI-Assisted Marine Risk Intelligence & Community Reporting Platform” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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Date: July 31, 2026

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to express my sincere appreciation to Dr. Manish Prateek, Director, MARS – Centre for Sustainable Community Development, for providing the opportunity to work as a GIS Engineer on this environmental intelligence project. I also extend my gratitude to the faculty and mentors of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous academic guidance and foundational training.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: AYUSH KUMAR

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INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP COMPLETION

Reference No.: MARS/INT-CERT/2026/065

Date of Issue: July 31, 2026

This is to certify that

Ayush Kumar

has successfully completed an internship with MARS – Centre for Sustainable Community Development from June 1, 2026 to July 15, 2026, contributing sincerely to the organisation's initiatives in sustainable community development. During this period, the intern demonstrated dedication, a strong learning attitude, and a genuine commitment to the values of sustainability and community service.

Sustainability is not a destination but a continuing journey. May the knowledge and experience gained during this internship inspire you to keep championing sustainable practices, and to carry this commitment forward into every community and career you touch in the years ahead.

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Marine heatwaves (MHWs) and abnormal sea-surface warming represent significant threats to global oceanic biodiversity, leading to mass coral bleaching events, reduced fish stock mobility, and the destabilization of coastal marine economies. Traditional monitoring systems often operate in isolated silos, requiring specialized oceanographic domain knowledge that restricts the ability of local conservation authorities and non-governmental organizations (NGOs) to act proactively.

The AI-Assisted Marine Risk Intelligence & Community Reporting Platform was conceptualized and developed during a 45-day technical internship with MARS – Centre for Sustainable Community Development. The primary objective was to build a cohesive, accessible platform that ingests raw satellite telemetry, processes sea-surface temperature anomalies against multi-decade climatology baselines, and models spatial risk zones using geographic information systems. Serving in the specialized role of GIS Engineer, my responsibility centered on resolving geographic projection and coordinate errors, implementing spatial boundaries, engineering cartographic symbology in QGIS, and delivering verified geospatial data layers for interactive web dashboard consumption.

4.2 Organization Profile

Name of the Organization: MARS – Centre for Sustainable Community Development

Corporate Identification / Domain: Environmental Data Systems, Community Sustainability & Ecological Informatics

Web & Communication: info@marsgroup.org.in | contact@marsgroup.org.in | marsgroup.org.in

Organizational Focus: MARS focuses on driving community-centric sustainability initiatives through technology-enabled monitoring. The organization develops field-level decision support tools, community hazard logging platforms, and ecological data

frameworks that empower grassroots stakeholders, academic researchers, and municipal environmental bodies.

4.3 Problem Statement

Unprecedented Marine Warming Trends: Rapid elevations in sea-surface temperatures require real-time tracking to prevent widespread marine ecosystem degradation.

Geospatial & Coordinate Incompatibilities: Raw satellite observation datasets often utilize continuous 0° to 360° longitude conventions. In standard GIS engines utilizing the WGS 84 datum (-180° to +180°), this creates critical boundary-wrapping bugs where points cross the international date line and distort global spatial layers.

Data Accessibility & Processing Gaps: Remote sensing files (NetCDF, HDF5) are multidimensional and computationally heavy, preventing non-technical field workers from visualizing localized thermal stress.

Isolated Field Reporting: Citizen reports regarding coastal pollution, fish kills, or algal blooms typically lack spatial verification against empirical satellite temperature grids.

4.4 Project Objectives

Objective 1: Ingest satellite telemetry from NOAA, OISST, NCEI, and OceanSODA data streams.

Objective 2: Design a spatial normalization pipeline to convert satellite coordinates into standard WGS 84 (EPSG:4326) format.

Objective 3: Calculate real-time sea-surface temperature anomalies (SST_Anomaly) against the 30-year 1991–2020 climatological baseline.

Objective 4: Model and visualize spatial heatwave stress using QGIS through custom graduated symbol configurations.

Objective 5: Define bounding corridors (isolating the Tropical Atlantic marine region) to eliminate unnecessary global computation.

Objective 6: Provide lightweight, verified vector data files and interactive plotting modules for frontend Streamlit dashboard integration.

4.5 Scope of the Project

The scope of this project encompasses spatial ETL (Extract, Transform, Load) pipelines, coordinate transformation algorithms, cartographic rendering, and web visualization hand-off. The geographic focus is bounded within the Tropical Atlantic corridor (Latitude: -20° to $+20^{\circ}$, Longitude: -60° to $+15^{\circ}$), an ecologically sensitive marine highway. The platform bridges raw remote-sensing data processing with actionable downstream GIS layers, supporting rapid identification of anomalous thermal zones without requiring users to run complex spatial software locally.

4.6 Technologies and Tools Used

Geographic Information Systems (GIS):

QGIS Desktop 3.x: Vector styling, projection remapping, print layout generation, and layer export.

Coordinate Reference System: EPSG:4326 (WGS 84 standard geographic coordinate reference).

Spatial Interpolation: Inverse Distance Weighting (IDW) and graduated vector binning.

Data Processing & Programming:

Python 3: Core programming language for coordinate correction scripts.

Pandas & NumPy: Tabular data processing, mathematical filtering, and anomaly computation.

Web Mapping & Frontend:

Plotly Express (plotly.express): Interactive Mapbox web plotting.

Streamlit: Python-based web framework for hosting the real-time marine risk dashboard.

Storage & Formats:

Comma-Separated Values (CSV), OGC GeoPackage (GPKG), and NetCDF raster structures.

4.7 System Architecture

The project follows a five-stage modular pipeline:

1. Data Ingestion — Satellite SST Telemetry (NOAA / OISST / NCEI / OceanSODA)
2. Geospatial ETL & Anomaly Core — Python Scripting: Longitude Realignment & SST Anomaly Math
3. GIS Engineering & Risk Modeling — QGIS Desktop (EPSG:4326): Spatial Bounding & Graduated Symbolology
4. Data Packaging & Database Store — GeoPackage / Clean CSV Spatial Grid Schema
5. Interactive Presentation Layer — Plotly Express & Streamlit Web Dashboard Engine

Ingestion Layer: Aggregates satellite measurements across global oceanic grids.

ETL & Normalization Engine: Corrects longitude wrapping bugs and isolates active thermal stress instances.

GIS Analysis Core: Applies symbology hierarchies to visualize thermal stress plumes without overlapping marker artifacts.

Storage Layer: Writes clean, permanent spatial tables to disk for downstream deployment.

Dashboard Layer: Renders dynamic web maps enabling user interactivity, zooming, and metadata inspection.

4.8 Methodology (GIS Engineer Responsibilities)

As the GIS Engineer, my technical workflow was divided into eight sequential engineering steps:

Step 1: Spatial Telemetry Diagnostics

Examined raw satellite datasets to evaluate coordinate conventions and data schema consistency. Identified that longitudinal data was provided in a continuous positive 0° to 360° tracking frame. When loaded into standard GIS tools alongside base map layers,

points in the Western Hemisphere plotted erroneously in outer space coordinates, fracturing the Atlantic basin.

Step 2: Algorithmic Coordinate Transformation

To align coordinates with the EPSG:4326 (WGS 84) coordinate reference system, I formulated and applied the following piecewise mathematical transformation:

$\text{Corrected_Lon} = \text{Longitude} - 360, \text{ if Longitude} > 180^\circ; \text{ else } \text{Corrected_Lon} = \text{Longitude}$

This programmatic transformation was integrated into the automated data pipeline, resetting longitudes exceeding 180° back into standard negative decimal degrees (e.g., $330.125^\circ \rightarrow -29.875^\circ$).

Step 3: Spatial Corridor Bounding

To prevent memory exhaustion during client-side dashboard rendering, I established a geographic bounding box isolating the Tropical Atlantic: Latitudinal Bounds: $-20^\circ \leq \text{Latitude} \leq +20^\circ$; Longitudinal Bounds: $-60^\circ \leq \text{Corrected_Lon} \leq +15^\circ$. This reduced extraneous global oceanic records while preserving 3,243 critical observation points across the study area.

Step 4: SST Anomaly Computation & Filtering

Thermal stress was isolated by evaluating measured temperatures against 30-year climatology baselines: $\text{SST_Anomaly} = \text{Current_SST} - \text{Average_SST}$. Records were filtered for active heatwave conditions where $\text{Risk_Label} \geq 1$ (isolating Watch, Warning, and Alert states), removing ambient background data.

Step 5: Advanced Symbolology & Cartographic Styling

Imported the sanitized dataset into QGIS Desktop under EPSG:4326. Resolved rendering errors where dense circular markers overlapped, creating illegible solid bars across the canvas. Replaced circular geometry markers with boundary-free square grids sized to match the satellite sensor's spatial resolution. Configured a Graduated Symbol Renderer based on SST_Anomaly, utilizing a high-contrast sequential color ramp (light pink to deep red) to visually isolate thermal plumes.

Step 6: Print Layout Design & Server Optimization

Designed formal cartographic map products using the QGIS Print Layout engine. Addressed Web Map Service (WMS) export failures (caused by server pixel dimension caps) by configuring tiled requests and adjusting export DPI parameters, ensuring crisp vector PDF outputs.

Step 7: Spatial Database Packaging

Prevented data loss by exporting temporary QGIS scratch layers into permanent OGC GeoPackage (.gpkg) and structured CSV formats, ensuring schema consistency across systems.

Step 8: Frontend Dashboard Hand-off

Authored Python integration modules using `plotly.express.scatter_mapbox` and `streamlit`. Provided the development team with pre-configured viewport parameters (centering at Latitude 0.0° , Longitude -22.5° with Zoom level 2.5), allowing the map to render interactively on the web dashboard.

4.9 Expected Outcomes

A mathematically verified, sanitized spatial dataset (`fixed_atlantic_heatwaves.csv`) free of projection and wrapping errors.

Publication-quality GIS vector cartography maps detailing marine heatwave distribution across the Tropical Atlantic corridor.

A reproducible coordinate normalization pipeline ready for automated production deployment.

Seamless integration of interactive spatial layers into the team's web-based environmental monitoring dashboard.

4.10 Certificates of Completion and Other Communication Mail Proof

Certificate Reference ID: MARS/INT-CERT/2026/065

Issued by: MARS – Centre for Sustainable Community Development

Date of Issue: July 31, 2026

Authorized Signatory: Dr. Manish Prateek, Director

(Scanned copy of Certificate and institutional correspondence attached below and in Appendix / Section 3, page iii).



CERTIFICATE OF INTERNSHIP COMPLETION

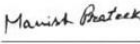
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CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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